

DATA ANALYSIS REPORT

AHJIN STUDIO'S BOX OFFICE SUCCESS BLUEPRINT

Analysis By: The Standard Deviants

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Overview

Ahjin Studios is entering the original film industry at a pivotal time. With legacy studios and streaming platforms locked in a high-stakes race to capture audience attention, the battle for box office dominance has never been more intense, or more data-driven. While compelling storytelling remains the heart of cinema, strategic, evidence-based decision-making has become its backbone.

This analysis fuses creative insights with financial intelligence, analyzing industry-wide patterns across genres, release timing, production budgets, critical and audience reception, and financial performance.

The goal: to decode the anatomy of a box office success, not just from an artistic standpoint, but through a financial and strategic lens.

To achieve this, we draw on five robust datasets from authoritative sources: IMDb, TMDb, Rotten Tomatoes, and The Numbers (for budget and gross earnings). Combined, they provide a comprehensive 360 degree view of the modern movie landscape, enabling Ahjin Studios to build a tactical playbook as it prepares to launch its first wave of original productions.

Problem Statement

Ahjin Studios is entering an increasingly volatile and saturated entertainment market without the cushion of existing IP, franchise power, or prior production history. The stakes are high. With production budgets reaching tens or even hundreds of millions, a single miscalculated decision can result in significant financial loss.

At the same time, audience preferences are evolving rapidly, driven by fragmented global tastes, streaming fatigue, and content oversaturation. The traditional formulas for success no longer guarantee results.

The central business question:

How can Ahjin Studios identify the most viable, high-impact film opportunities, without relying on established IPs or celebrity-driven marketing, while optimizing for ROI and long-term growth?

Business Objectives

1. **Identify commercially viable film characteristics:** Analyze which genres, runtimes, and themes consistently perform well across different markets.
2. **Optimize budget-to-return strategies:** Uncover patterns in production cost vs. box office return to guide investment decisions and reduce financial risk.
3. **Pinpoint strategic release timing:** Examine seasonal and monthly trends to determine when original films are most likely to succeed.

Data Sources

To ensure a comprehensive and multidimensional analysis of the film industry, this analysis draws from six authoritative datasets sourced from major platforms.

1. **bom.movie_gross.csv — Box Office Mojo:** This dataset provides domestic box office gross revenue figures, allowing us to track the financial performance of films once they hit theaters.
2. **im.db — IMDb:** A robust SQLite database containing detailed information on movies, ratings, cast, crew, and production details.
3. **rt.movie_info.tsv — Rotten Tomatoes:** This dataset offers high-level movie metadata, including critic scores, genre tags, and release years. It provides valuable contrast to IMDb's user ratings by introducing aggregated critic sentiment.
4. **rt.reviews.tsv — Rotten Tomatoes:** A deeper layer of the Rotten Tomatoes data, this dataset consists of individual reviews written by critics, along with binary ratings (fresh/rotten) and publication sources.
5. **tmdb.movies.csv — The Movie Database (TMDb):** This metadata-rich dataset includes film popularity metrics, genre information, and release dates.
6. **tn.movie_budgets.csv — The Numbers:** This dataset supplies detailed financial breakdowns, including production budgets, domestic and worldwide gross, and release dates. It is critical for calculating return on investment (ROI) and understanding the financial stakes of various film categories, budgets, and release strategies.

Constraints

- The latest dataset is recorded as at 2020, which is 5 years in the past.

Data Understanding

Dataset Overview

- **The Numbers:** Emphasizes *economic performance*. Includes: production budgets, domestic grosses, and global revenue.
- **Box Office Mojo:** Highlights *regional box office split* between domestic (U.S./Canada) and international markets, along with studio attribution.
- **TMDb (The Movie Database):** Offers *popularity indicators* (e.g vote counts, genre tags) drawn from a wide user base.
- **Rotten Tomatoes:** Centers on *critical reviews*, freshness scores, maturity ratings, and editorial metadata.
- **IMDb:** Focuses on *audience reception*, including user ratings, genre tags, and runtime information.

Each dataset enriches our understanding of film dynamics by capturing a different angle; economic, critical, cultural, or popular.

1. The Numbers (Financial Performance)

- **id:** Internal or scraped identifier for the film (not officially from The Numbers).
- **release_date:** Official U.S. theatrical release date of the film.
- **movie:** Full title of the film.
- **production_budget:** Estimated cost of producing the film in USD.
- **domestic_gross:** Total box office revenue from U.S. theaters (in USD).
- **worldwide_gross:** Combined domestic and international box office revenue (in USD).

2. Box Office Mojo (Studio and Revenue Split)

- **title:** Official name of the film.
- **studio:** Abbreviated name of the distributing or producing studio (e.g DIS for Disney).
- **domestic_gross:** Total theatrical earnings in the U.S./Canada (North America).
- **foreign_gross:** Total earnings from theaters outside North America.
- **year:** Year of the film's U.S. release.

3. TMDb (Popularity and User Voting)

- **genre_ids:** List of numeric genre identifiers corresponding to the film's genres.
- **id:** Unique identifier assigned by TMDb.
- **original_language:** ISO 639-1 code representing the movie's original language (e.g en, fr).
- **original_title:** Original title in the native language/script of the film.
- **popularity:** TMDb's proprietary popularity score based on user interactions.
- **release_date:** Date the film was officially released.

- **title:** Localized or translated title used for international or platform display.
- **vote_average:** Average user rating on a scale from 0 to 10.
- **vote_count:** Total number of votes submitted by TMDb users.

4. Rotten Tomatoes (Critic Reviews and Sentiment)

- **id:** Unique identifier for the movie or review entry.
- **review:** Full-text review provided by a film critic.
- **rating:** The critic's score (e.g 3.5/5, B+).
- **fresh:** Sentiment label assigned to the review (e.g fresh or rotten).
- **critic:** Name of the critic who wrote the review.
- **top_critic:** Flag indicating whether the critic is a recognized top critic.
- **publisher:** Media outlet or publication the critic represents.
- **date:** Date the review was published.
- **synopsis:** Short plot summary of the film.
- **maturity_rating:** Age-based rating (e.g PG-13, R).
- **genre:** Primary genre(s) assigned to the film.
- **director:** Director(s) of the film.
- **writer:** Writer(s) or screenwriters of the film.
- **theater_date:** Official U.S. theatrical release date.
- **dvd_date:** Home media (DVD) release date.
- **currency:** Currency used for box office values (typically USD).
- **box_office:** Total theatrical revenue (not split by region).
- **runtime:** Duration of the film in minutes.
- **studio:** Producing or distributing studio's name.

5. IMDb (User Ratings and Content Metadata)

- **movie_id:** Unique IMDb identifier (typically the tconst format).
- **primary_title:** Main title under which the film is listed or released.
- **original_title:** Title in the film's native or original language.
- **start_year:** Year the film was released or premiered.
- **runtime_minutes:** Duration of the film in minutes.
- **genres:** Comma-separated list of genre labels (e.g Drama, Action).
- **averagerating:** Average IMDb user rating on a scale from 1 to 10.
- **numvotes:** Number of user ratings submitted.

Data Preparation

1. Managing Missing Data

- Essential fields like genre, runtime, content rating, and release dates were dropped if missing as they're crucial for analysis.

- For non-essential categorical fields (e.g writer, director, studio) missing values were filled with "Unknown" to preserve the dataset size.
- Numerical fields such as runtime and revenue figures were imputed using the median to avoid skewing the distribution.

2. Converting Data Types

- Monetary fields (production_budget, domestic_gross, worldwide_gross, box_office) were cleaned by stripping out symbols (e.g \$) and converted to float for analysis.
- Date columns were converted to proper datetime format, allowing extraction of time-based features like year, month, weekday, and season.

3. Handling Outliers & Duplicates

- Outliers in revenue and runtime were retained as they reflect valid blockbuster or niche films.
- Duplicate records, especially in TMDb and Rotten Tomatoes, were identified and removed to maintain data integrity.

4. Feature Engineering

- Genre fields were standardized:
 - TMDb genre IDs were mapped to readable labels.
 - Genre strings from IMDb and Rotten Tomatoes were split and cleaned for consistency across datasets.
- Box office categories (e.g Flop, Low, Mid, Blockbuster) were created to group films by revenue.
- Runtime ranges were also bucketed to explore trends based on movie length.

5. Dataset Integration

- Rotten Tomatoes reviews were combined with metadata using a left join on movie ID.
- IMDb movie basics were merged with rating data to unify structural and audience metrics.

6. Final Sanity Checks

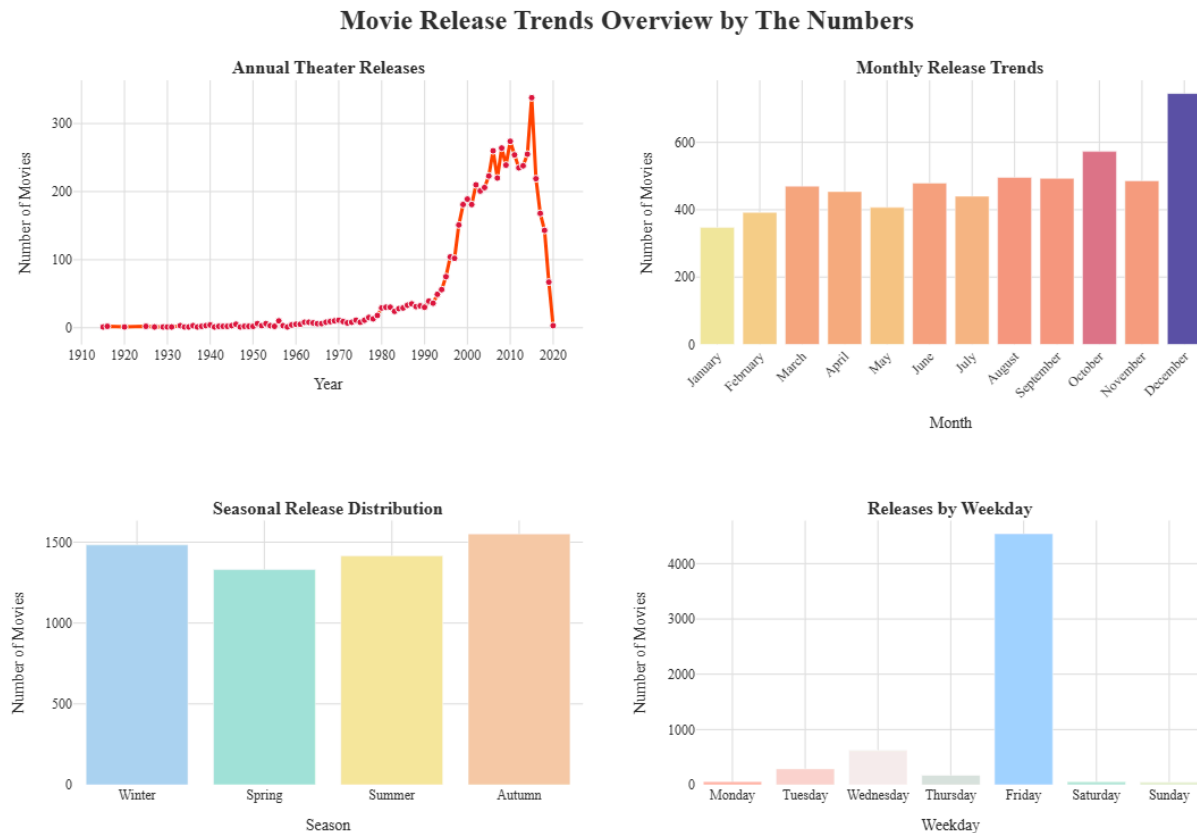
Before moving to analysis, all datasets were validated for:

- Remaining missing values
- Duplicate records
- Correct data types across all columns

Exploratory Data Analysis

Time Series Analysis

1. The Numbers dataset



The figure above shows observations on theatrical release timing as recorded by The Numbers.

Decade-Level Trends

- There's a clear upward trajectory in theatrical releases starting from the 1980s.
- The 1990s, 2000s, and 2010s each saw exponential growth, with over 2,000 films released in both the 2000s and 2010s.
- This surge aligns with global market expansion, the rise of independent studios, and lower barriers to production and distribution.

Monthly Seasonality

- December dominates with the highest number of releases, leveraging holiday audiences and positioning for awards.

- October, August, and September show strong activity, likely targeting horror fans, end-of-summer viewers, and fall festival circuits.
- January and February remain the quietest months, often used for lower-stakes releases or clearing studio pipelines.

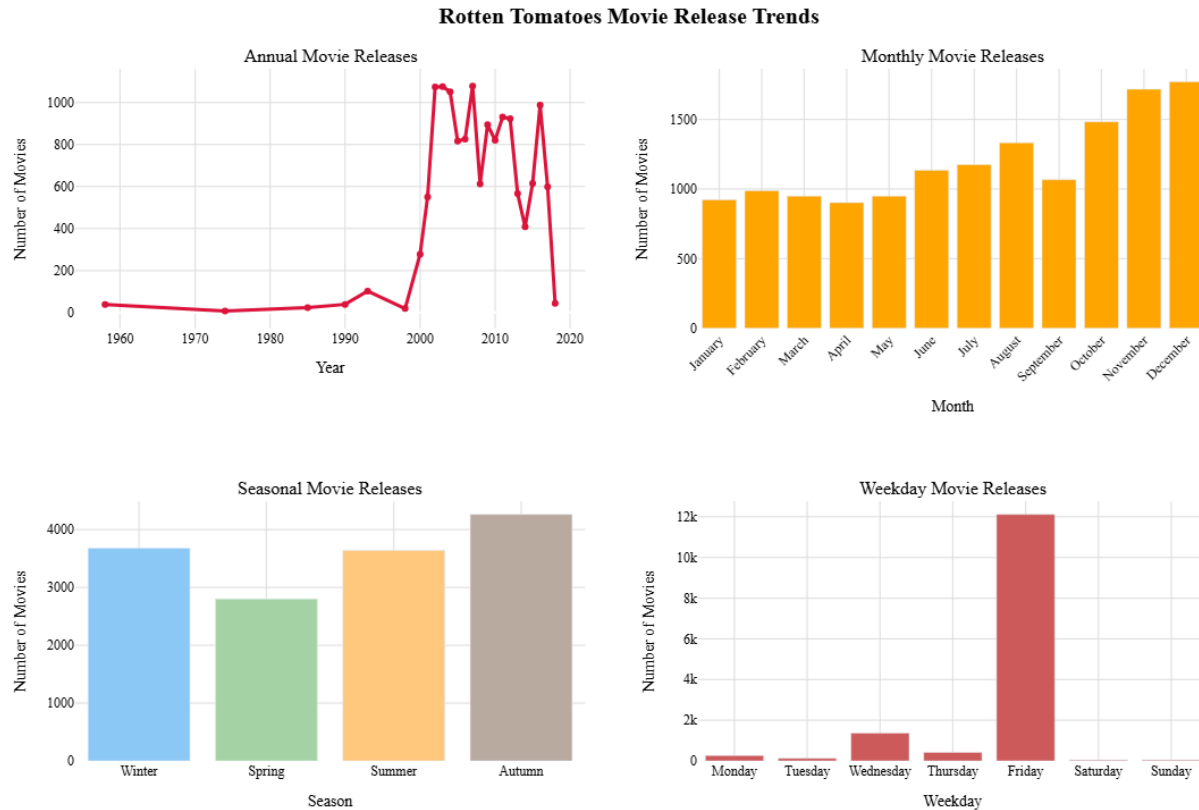
Seasonal Patterns

- Autumn and Winter lead in theatrical output, likely benefiting from audience availability, holiday traffic, and awards buzz.
- Spring lags behind, indicating a less competitive window for major studio launches.

Weekday Release Strategy

- Friday is the preferred release day by a wide margin, with over 4,500 films debuting, optimized for weekend box office performance.
- Tuesday and Wednesday follow, often used for limited or platform releases and mid-week press attention.
- Monday, Saturday, and Sunday releases are rare, as they offer weaker media cycles and reduced promotional impact.

2. Rotten Tomatoes



The figure above shows observations on theatrical release timing as recorded by Rotten Tomatoes

Decade Trends

- There has been a significant rise in the number of theatrical releases over the decades, particularly from the 1980s onward.
- The 2000s marked the peak, with over 10,000 films released – a period that likely reflects the boom of both mainstream cinema and independent productions.
- A slight decline in the 2010s (7,787 films) may be attributed to either data reporting lags or the increasing shift toward streaming platforms, especially post-2015.

Monthly Patterns

- December (3,154 releases) leads in volume, followed closely by November (3,023) and January (2,537), aligning with holiday season marketing pushes and awards season eligibility windows.
- Summer months, June, July, and August, also record high numbers, typically serving as launchpads for blockbusters and family-oriented films.

- February sees the fewest releases (1,569), consistent with its historical placement as a slower post-awards season month.

Seasonal Patterns

- Autumn (7,643 releases) and Winter (7,260) dominate theatrical schedules, reinforcing their strategic value for both holiday audiences and awards contention.
- Spring (5,591 releases) consistently shows the lowest volume, suggesting it is considered less commercially favorable for major releases.

Weekday Patterns

- Friday is the most common release day by a large margin, with 20,048 films, reaffirming its central role in maximizing opening weekend performance.
- Wednesday (3,402 releases) is the second most frequent, often used for limited releases or midweek festival launches.
- Sunday and Saturday have the fewest releases, likely due to reduced media buildup and minimal marketing momentum.

Cross-Platform Summary of Theatrical Release Timing

Decade-Level Trends

- Both datasets show a clear upward trend in theatrical releases beginning in the **1980s**, peaking in the **2000s**.
- **The Numbers** indicates exponential growth across the 1990s–2010s, driven by globalization, indie filmmaking, and digital distribution.
- **Rotten Tomatoes** shows the 2000s as the dominant decade (10,000+ films), with a slight dip in the 2010s, possibly due to streaming and incomplete records.

Monthly Seasonality

- **December** is the top release month across both platforms, capitalizing on holiday attendance and awards-season visibility.
- **October and November** also see strong activity, aligning with horror releases, fall festivals, and year-end prestige campaigns.
- **February** is consistently the quietest month, serving as a dumping ground for underperformers or niche titles.

Seasonal Patterns

- **Autumn and Winter** dominate release volumes on both platforms, ideal for awards strategies and holiday viewership spikes.
- **Spring** is the weakest season overall, signaling its reduced strategic value for studios.

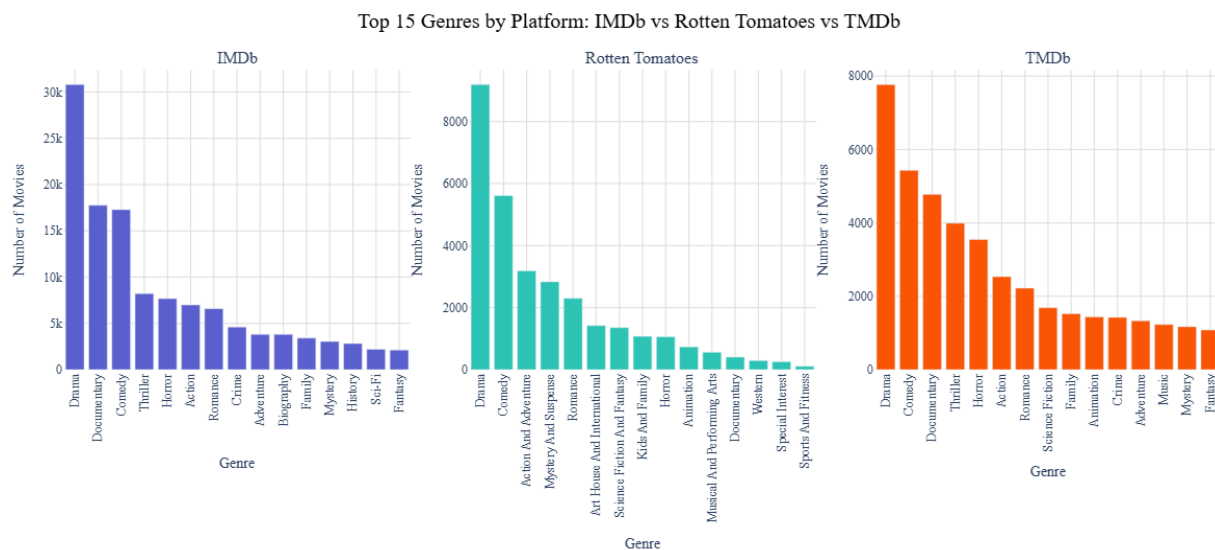
- **Summer** (June–August) is stronger on Rotten Tomatoes, suggesting heavier tracking of blockbusters and youth-oriented content.

Weekday Strategy

- **Friday** is overwhelmingly the most popular release day across both datasets — optimized for opening weekend momentum.
- **Wednesday and Tuesday** follow, often linked to platform releases or midweek drops for media buzz.
- **Saturday, Sunday, and Monday** remain rare release days, reflecting their poor alignment with press cycles and ticket-sale build-up.

Univariate Analysis

1. Genre Distribution



The figure above shows the distribution of different genres in IMDb, Rotten Tomatoes and TMDb

Analysis of Top 15 Genres Across Platforms

IMDb: *Drama* significantly leads the genre distribution, boasting over 30,000 titles, more than twice the count of any other genre. *Documentary* and *Comedy* follow, indicating a balance between narrative fiction and non-fiction content. Genres such as *Thriller*, *Horror*, and *Science Fiction* are clearly delineated.

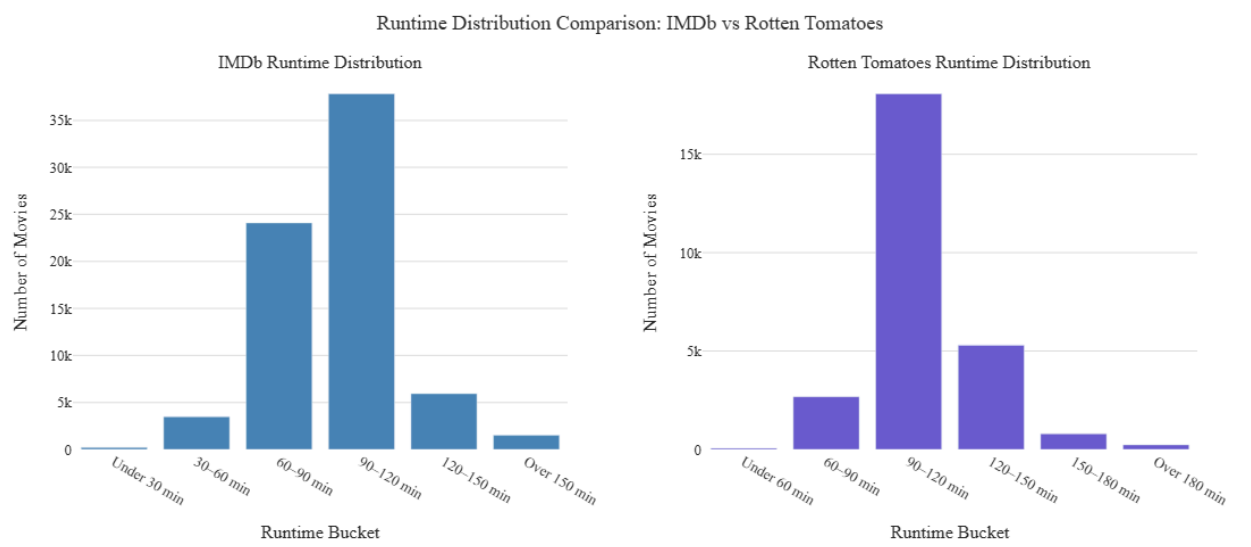
Rotten Tomatoes: While *Drama* and *Comedy* also top Rotten Tomatoes, the platform’s categorization style diverges. Instead of strict industry-standard genres, the system often uses editorially-driven labels such as “*Mystery and Suspense*” or “*Action and Adventure*.” It also includes thematic groupings like “*Art House and International*” or “*Special Interest*,” likely aimed at guiding viewer expectations rather than reflecting production-side classifications. The notably low representation of *Documentary* may be due to selective review criteria or differing inclusion thresholds.

TMDb: *Drama*, *Comedy*, and *Documentary* are the most frequent genres. The relatively high number of Documentaries suggests the platform’s inclusivity toward independent, international, and community-submitted factual content.

Cross-Platform Summary

Across all three platforms, *Drama* and *Comedy* consistently emerge as the most prominent genres, underscoring their universal appeal and market saturation.

2. Runtime Distribution



The figure above shows runtime comparison of films from IMDb and Rotten Tomatoes

Runtime Realities: Strategic Runtime Insights for Ahjin Studios

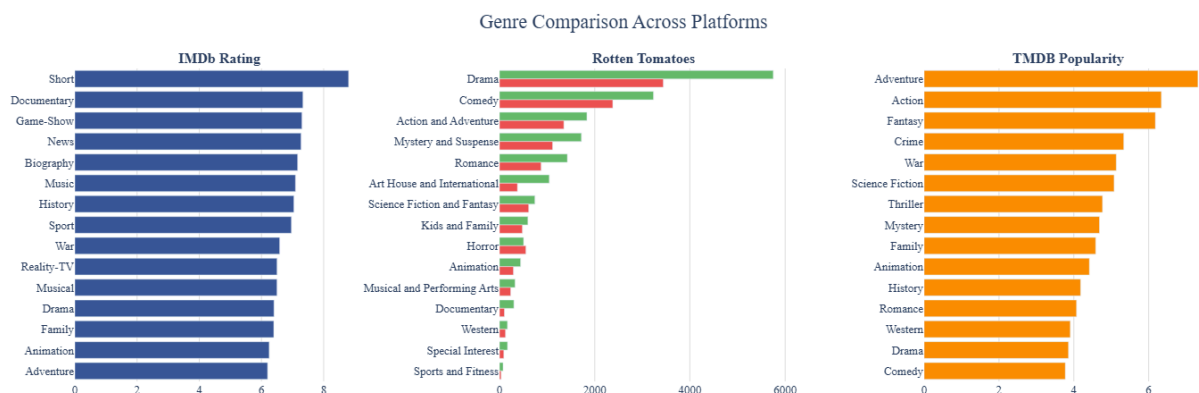
Key Findings from Runtime Analysis

- The 90–120 minute range is the clear industry standard across platforms:

- IMDb: 37,832 films fall within this range, the most of any category.
- Rotten Tomatoes: 18,073 films in this bracket, also the most dominant.
- 60–90 minute films have a strong presence on IMDb (24,099 titles), indicating frequent production, particularly for independent or lower-budget works. However, their lower count on Rotten Tomatoes suggests limited critical traction.
- Extremely short (<60 minutes) and long-form (>150 minutes) films are niche categories. These formats tend to see limited critical or commercial recognition, making them higher-risk for mainstream production strategies.
- IMDb’s inclusion of a large volume of short films (under 30 and 30–60 minutes) points to experimental and indie trends. While valuable for talent scouting or platform-specific content, they are less applicable to Ahjin’s core theatrical and commercial objectives.

Bivariate Analysis

1. Cross Platform Genre Popularity Comparison



The figure above shows cross platform genre popularity comparisons

Observations

1. Quality Perception (IMDb Ratings)

- **Shorts** (avg. 8.8) and **Documentaries** top critical acclaim but have limited commercial depth.
- **Drama**, **Biography**, and **History** offer high ratings with narrative strength.
- **Animation** and **Adventure** score lower (~6.2) but serve spectacle and family appeal.

2. Critic Sentiment (Rotten Tomatoes Fresh vs Rotten)

- **Drama** and **Comedy** lead in volume but show critical splits.
- **Mystery/Suspense, Romance, and Action** skew fresh, broad critic approval.
- **Horror** and **Family** genres are hit-or-miss.
- **Documentaries** consistently score positive reviews.

3. Viewer Popularity (Audience Interest)

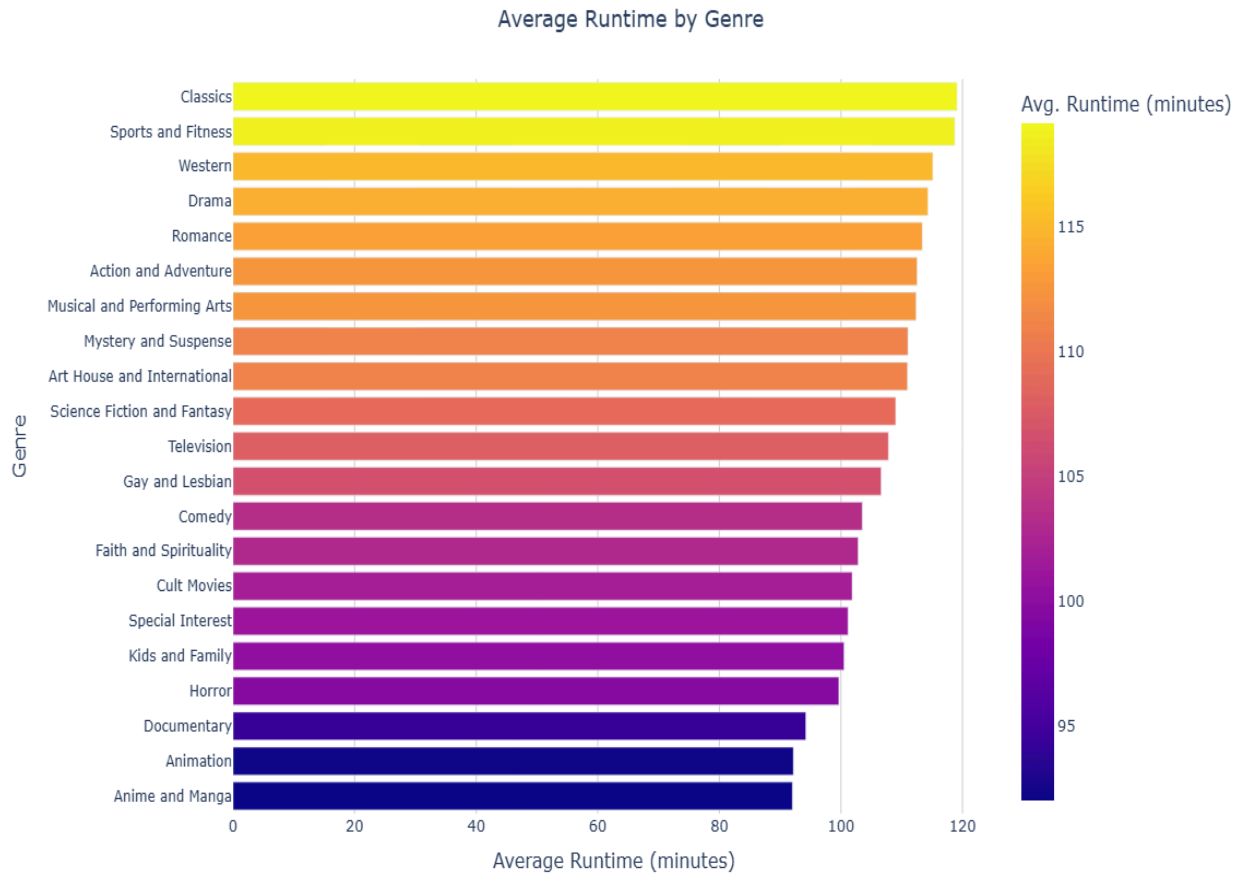
- **Adventure** and **Action** dominate viewer demand, great for box office and streaming.
- **Fantasy, Crime, and War** carry strong followings.
- **Drama** is common but less buzzy.
- **Sci-Fi, Animation, and Family** maintain targeted fanbases.

Takeaways

1. **Prioritize high-impact genres:** Action + Adventure for reach; Biography + Mystery for prestige.
2. **Avoid overdone categories:** Drama requires uniqueness to cut through noise.
3. **Low-budget, high-rating opportunities:** Documentaries and Suspense genres offer ROI potential.
4. **Align audience + critic taste:** Genres with dual acclaim (e.g Biography, Mystery) are safest bets.

2. Average runtime by genre

1. Rotten Tomatoes



The figure above examines the relationship between movie genre and average runtime from Rotten Tomatoes.

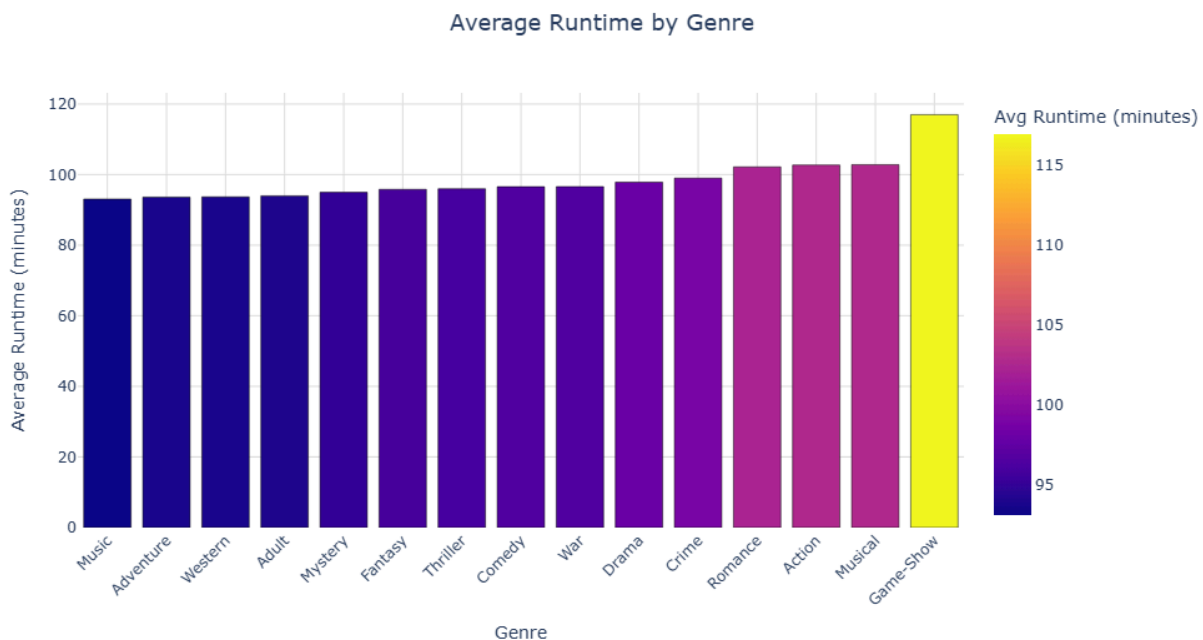
Observations

Longest Average Runtimes:

- Game-Show: 117.0 minutes
- Musical: 102.8 minutes - Extended duration for song sequences and choreography
- Biography: 100.4 minutes - Complex life narratives require substantial storytelling time
- History: 99.8 minutes - Period detail and historical context demands
- War: 99.6 minutes - Epic scope and battle sequences extend duration
- Commercial Sweet Spot (90-96 minutes)
- Drama: 96.4 minutes - Balanced storytelling within commercial viability
- Crime: 96.2 minutes - Complex plots with character development
- Romance: 94.8 minutes - Relationship development timeframe

- Thriller: 94.5 minutes - Tension building within attention limits
- Animation: 86.9 minutes - Family-friendly attention span considerations
- Comedy: 89.2 minutes - Quick pacing for comedic timing optimization
- Horror: 91.1 minutes - Concentrated intense experiences work best

2. IMDb



The figure above shows the distribution of genre runtimes according to IMDb

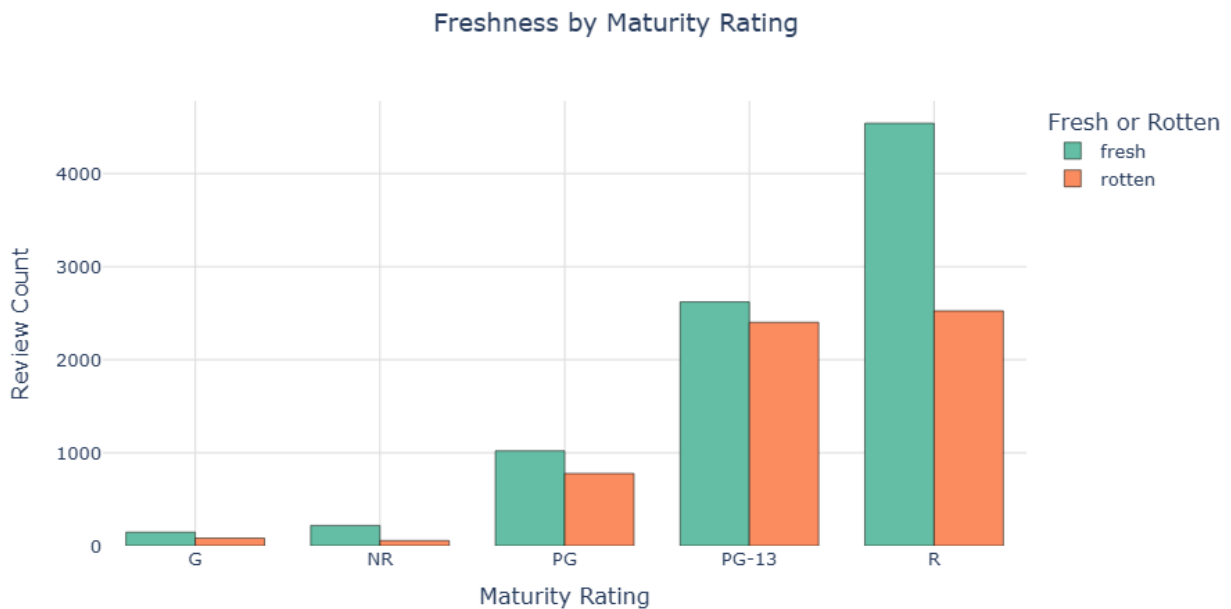
Observations:

- **Game-Show** tops the chart with the longest average runtime at **117 minutes**
- **Musical** and **Action** genres trail closely, averaging around **103 minutes**. Their longer runtimes likely accommodate elaborate sequences, musical numbers in one case, and choreography or effects-heavy scenes in the other.
- Genres like **Romance**, **Crime**, and **Drama** hover between **97 to 102 minutes**, reflecting the narrative depth and character development these categories typically demand.
- **Comedy** lands at a compact **96.6 minutes**, balancing tight plot pacing with enough room for comedic buildup and payoff.
- **Fantasy**, **Mystery**, and **Thriller** cluster tightly in the **95–96 minute** range, possibly a deliberate choice to sustain tension and maintain audience engagement without overstaying their welcome.

- **Adult** films clock in at **94 minutes** on average, consistent with the genre's straightforward, often single-threaded structure.
- At the shorter end, **Adventure, Music, and Western** genres average around **93–94 minutes**. These may lean on visually-driven storytelling or leaner plots that don't require extended runtimes.

Bottom line: Genres with high production complexity or multi-layered narratives (think Game-Show, Action, Musical) tend to run longer. In contrast, genres built around atmosphere, pace, or simplicity (like Thriller or Western) favor tighter durations for impact.

2. Maturity Ratings Freshness



The figure above shows the freshness of maturity ratings according to Rotten Tomatoes critic scores.

Observation

- **G** → 74% Fresh (489/662): Smallest output, but impressively well-received.
- **PG** → 64% Fresh (2,790/4,366): Family-friendly with a decent critical nod.
- **PG-13** → 54.5% Fresh (4,095/7,512): Dominates volume, splits opinions.
- **R** → 63.7% Fresh (8,137/12,773): High output, strong critical favor.
- **NR** → 78.6% Fresh (1,436/1,826): Highest approval, likely art-house or niche.

Insight

- **NR** reigns with critics – possibly due to creative freedom.

- **PG-13** saturates the space but lacks consistent acclaim.
- **R** hits a sweet spot: popular and respected.
- **G/PG** play it safe – generally liked, rarely loved.

3. Box Office Performance Comparison Between Theater and DVD Monthly Releases



The figure above the comparison of box office performance of Theater and DVD monthly releases

Observation:

Theater Releases

- **Blockbusters (\$100M+)**: Peak in **June, November, and December** – classic summer and holiday blockbuster slots.

- **High-grossers (\$50M–100M):** Strong in **Q4**, especially **December** — aligns with **awards season and holiday family crowds**.
- **Mid-tier (\$10M–50M):** Year-round presence with **notable spikes in August and October** — great for **crowd-pleasers, horror, or genre hits**.
- **Low-tier (\$1M–10M):** Steady flow throughout the year with blips in **March, May, and December** — reflects their **long-tail appeal**.
- **Flops (<\$1M):** Modest clustering in **April–June** and **October**. Likely **festival leftovers or misfires**.

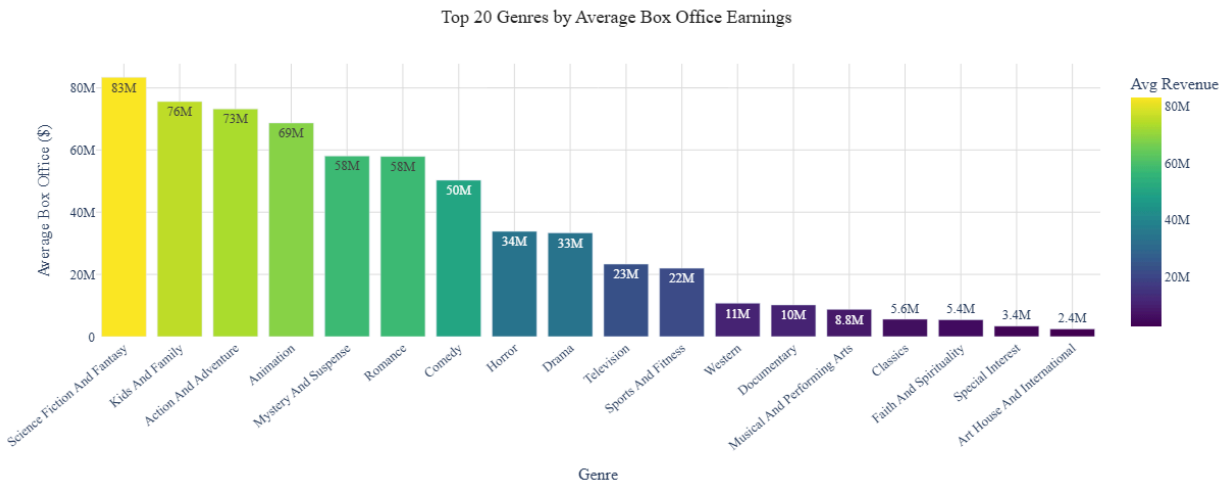
DVD Releases

- **Blockbusters:** Hit shelves mostly in **Feb–March** and **Nov–Dec** — aligns with **Q1 retail and holiday gift windows**.
- **High-grossers:** Similar rhythm, with some releasing a touch earlier.
- **Mid & Low-tier:** Maintain a **steady year-round drip**, signaling long-tail home-viewing relevance.
- **Flops:** Drop early **Q1 (Jan–March)**. Likely **closing the loop** on a failed theatrical run.

Insight

- **Q4 (Oct–Dec)** is the **golden corridor**. Ideal for blockbusters, Oscar bait, and high-volume family titles.
- **Q2–Q3 (May–Aug)** are **prime for mid-budget hits** for action, horror, comedies, or youth-driven films.
- **DVD releases trail theaters by ~3–4 months**, reinforcing traditional windowing patterns.
- **Avoid rolling out risky indies or niche titles in April–June**. Flop clusters suggest that stretch is an uphill battle.

4. Average Box Office of the Top 20 Movie Genres

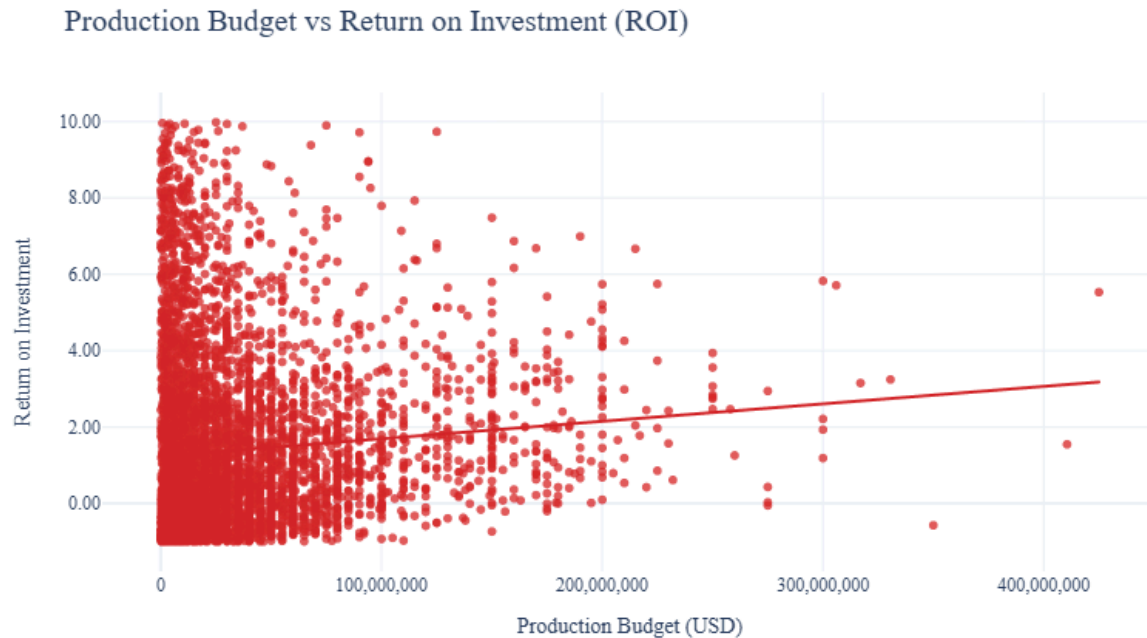


The figure above shows the top 20 box office performers by genre

Observation

- **Science Fiction & Fantasy tops all genres in average box office revenue**, closely followed by **Kids & Family, Action & Adventure, and Animation**. These genres consistently generate strong commercial returns thanks to their wide audience reach, franchise-building opportunities, and merchandising potential.
- Mid-performing genres like **Mystery & Suspense, Romance, and Comedy** achieve solid results, supported by their relatability and seasonal appeal, though they don't match the earnings of top-tier categories.
- While **Drama and Horror** enjoy strong popularity, they tend to produce lower average revenue. Their success often depends more on critical reception or viral momentum than on broad market appeal.
- **Niche genres, including Documentary, Musical, Faith-based, Western, and Art House, show the lowest average box office returns**, largely due to limited theatrical runs and more specialized target audiences.

5. Production Budget vs Return on Investment (ROI): Are Bigger Budgets Better?



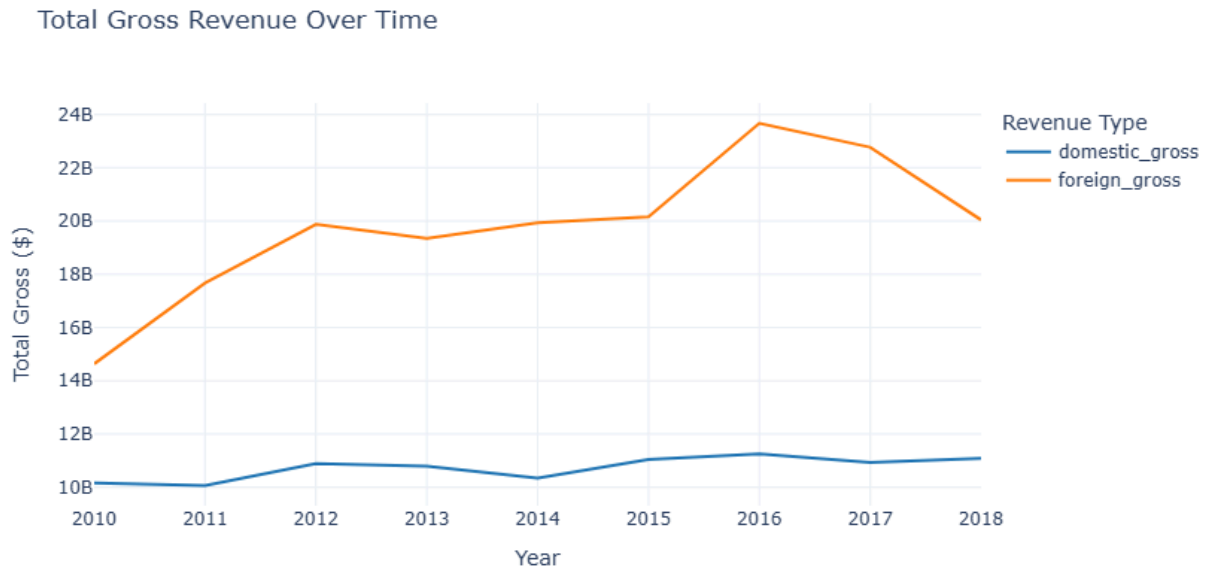
The figure above compares film production budget and their ROI

Observation

- **Spending more on production doesn't ensure higher returns.** The data reveals almost no correlation between a film's budget and its return on investment (ROI), meaning larger budgets do not consistently translate into greater profitability.
- **Low-budget films (under \$10M) stand out as the most efficient, delivering the highest ROI across the board.** These projects often outperform both mid-range and big-budget films in terms of profit relative to cost.
- Mid-budget films (\$10M–\$30M) show no statistical ROI advantage over high-budget films, suggesting that this tier lacks clear financial leverage.
- Although high-budget films (above \$70M) generate higher total box office gross, their ROI tends to be lower. The larger revenue is often offset by massive production and marketing expenses.
- Critically, the median ROI across all films is below 1, meaning that over half of all films fail to earn back even twice their production cost.

Multivariate Analysis

1. Total Gross Revenue Over Time (2010–2018)



The figure above shows comparison of domestic and foreign gross trends over time.

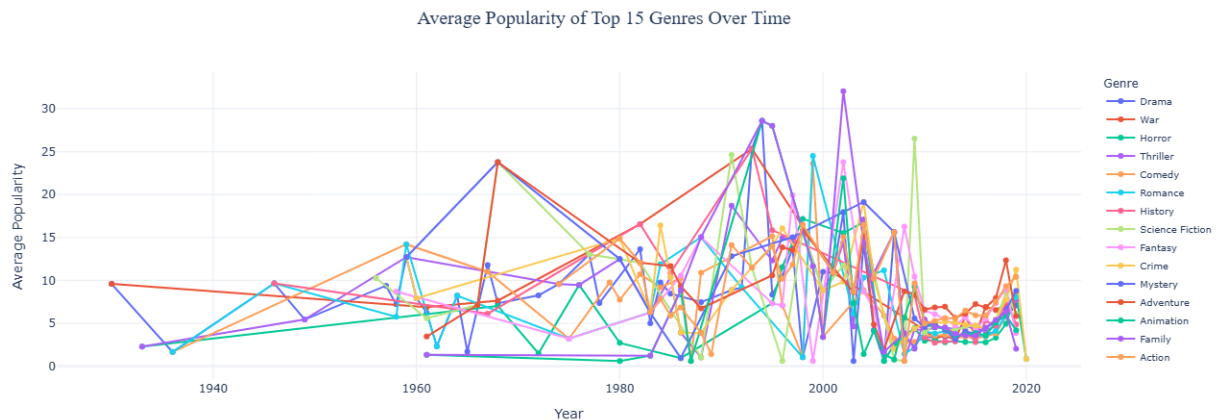
Observation:

- Both domestic and foreign revenues have seen growth over time, with foreign gross consistently outpacing domestic year over year.
- The largest total grosses were seen between 2015–2018, reflecting a rise in international box office dominance during that period.

Recommendation:

Ahjin Studios should target international markets, track yearly performance trends, and develop films with universal resonance using this data to support both creative and financial decision-making.

2. Genre Popularity Over Time



The figure above is a visual representation of genre popularity over time.

Observation:

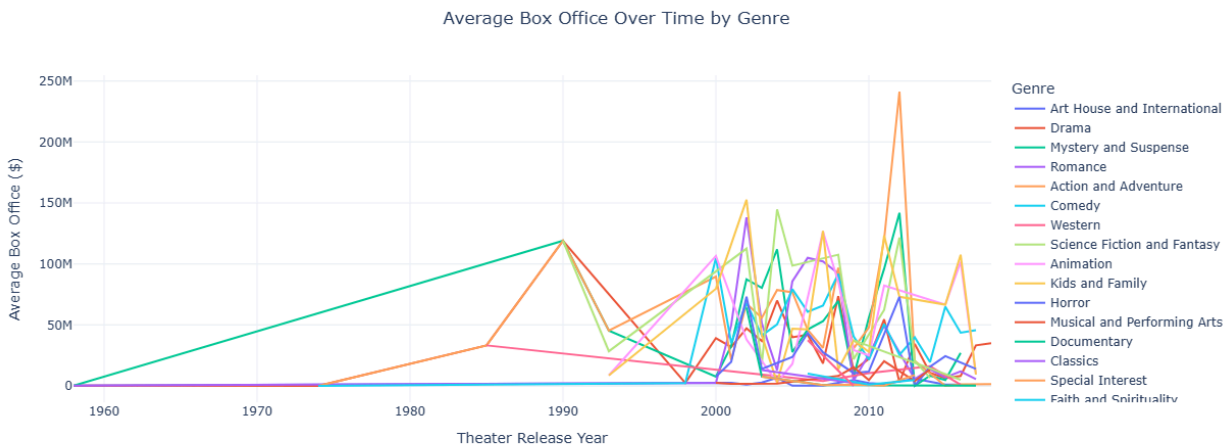
- **The Early Days (1930s–1960s):** Genres like War and Romance held steady ground, showing modest but consistent popularity. This reflects the more centralized studio system of the time and limited global distribution.
- **Genre Flux (1970s–1990s):** As cinema diversified, genres like Thriller, Horror, and Adventure gained traction. Peaks and valleys reflect the rise of blockbuster experimentation. No single genre dominated – trends were short-lived, influenced by cultural and tech shifts.
- **The Explosion Era (2000s):** A dramatic spike in popularity occurred around major franchises like Harry Potter and The Lord of the Rings. Animation and Sci-Fi also rose with advancements in CGI. Comedy and Drama became more volatile in audience reception.
- **The Decline (2010s–2020s):** Despite high production volumes, average popularity across genres dipped. Streaming may be the cause – niche content thrives but mass popularity is harder to achieve. Genre loyalty has faded; audiences now mix and match across genres.

Insights:

- **Adapt to Trend Volatility:** Genre preferences are no longer predictable. Every genre has a moment, but none dominate for long.
- **Flexible Production Planning:** Avoid overreliance on any single genre. Instead, embrace genre experimentation and blending.

- **Audience Engagement:** Use micro-trends and data to guide short-term content investments. Niche success is often more realistic than mass appeal in the modern landscape.

3. Genre Performance at the Box Office Over the Years



The figure above shows the average performance of various genres at the box office

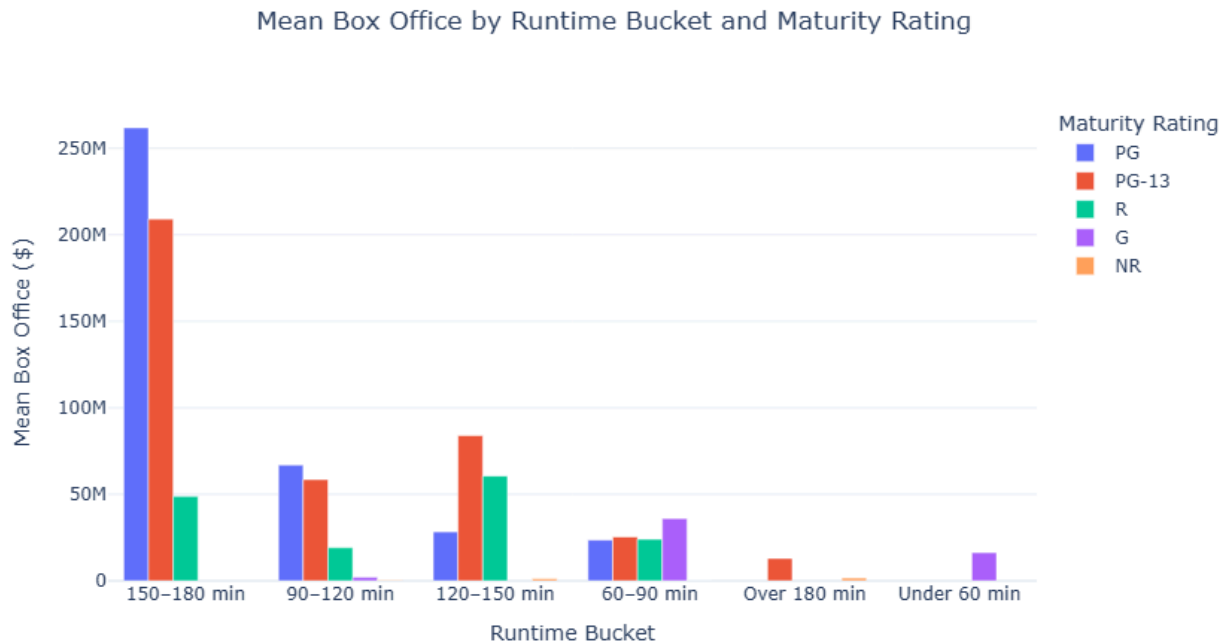
Observation:

- **High Volatility:** Nearly all genres show inconsistent performance, especially after 1990.
- **Top Performers:** Action and Adventure, Sci-Fi and Fantasy, and Animation dominate with the highest revenue peaks.
- **Low Performers:** Documentary, Special Interest, and Faith & Spirituality consistently earn lower due to niche audiences.

Insights

- **Revenue Forecasting:** Be cautious, average genre performance can be misleading due to spikes.
- **Risk Management:** Pair high-risk blockbusters with steady performers in low-variance genres.
- **Portfolio Diversification:** Explore niches for consistent low-cost returns while chasing occasional blockbusters.

5. Runtime & Maturity Rating vs Box Office Performance



The figure above shows the relationship between film runtimes and their maturity ratings against box office performance.

Observation:

- **Films in the 150–180 min runtime range with PG and PG-13 ratings have the highest mean box office, earning well above \$200M.**
- R-rated and PG-13 films in the 120–150 min range perform moderately well, averaging between \$50M–\$100M.
- Short Films Underperform: Runtimes below 90 minutes, especially under 60 min, earned significantly less, with many maturity ratings showing minimal revenue.
- Under 60 min & Over 180 min: These two extremes see the lowest box office across all maturity categories, indicating limited commercial appeal.
- G-rated films: Generally underperformed across all runtime buckets except the 60–90 min category.

Insights:

- **Optimal Runtime Sweet Spot:** Target runtimes between 120–180 minutes, especially when creating PG or PG-13 rated films to maximize reach and revenue.
- **Family-Friendly Franchises:** PG-rated films, especially long ones, hold immense commercial potential ideal for franchises and holiday releases.

- **Avoid Extremes:** Avoid producing films under 60 minutes or over 180 minutes unless for niche or artistic purposes.
- **Rating Matters:** Choose ratings carefully, G-rated films may be too narrow in appeal, while R-rated ones limit audience size. PG-13 offers the best mix of depth and accessibility.

Hypothesis Testing

Hypothesis 1: PG/PG-13 Films Earn More Revenue Than G or R-Rated Films

Business Context:

It's commonly believed that PG and PG-13 ratings help broaden audience appeal, leading to higher box office earnings.

Test Details:

- Method: ANOVA
- Groups Compared: G, PG, PG-13, R
- Measured Variable: Average box office revenue

Results:

- $p < 0.001$ (highly significant)
- Mean Revenues:
 - PG-13: \$65.8M
 - PG: \$64.4M
 - G: \$30.2M
 - R: \$29.5M

Interpretation:

There is a statistically significant difference in revenue across maturity ratings. PG and PG-13 films generate over twice the revenue of G- or R-rated titles.

Recommendation:

Ahjin Studios should prioritize films rated PG or PG-13 to maximize commercial reach and earnings.

Hypothesis 2: Mid-Budget Films Yield Higher ROI Than High-Budget Productions

Business Context:

It is often assumed that films with mid-level budgets (\$10M–\$30M) strike a balance between cost and profit, delivering better ROI than big-budget productions.

Test Details:

- Method: OLS Regression (budget tier as categorical variable)
- Budget Categories:
 - Low (<\$10M)
 - Mid (\$10M–\$30M)
 - High (>\$30M)
- Measured Variable: Return on Investment (ROI)

Results:

- Mid-budget vs High-budget: No significant ROI difference ($p = 0.649$)
- Low-budget vs High-budget: Low-budget films significantly outperform, with ROI averaging 4.2× higher ($p < 0.001$)

Interpretation:

Mid-budget films do not show a statistical advantage over high-budget ones in terms of ROI. However, low-budget films deliver the strongest returns.

Recommendation:

Ahjin Studios should focus on low-budget projects for higher financial efficiency and reduced risk. Mid-budget productions should only be pursued if they offer clear market or creative advantages.

Conclusion

These hypothesis tests reveal that maturity rating significantly influences box office revenue, with PG and PG-13 films performing best.

Meanwhile, budget tier impacts ROI only at the low end, where low-budget productions clearly outperform. These insights should directly inform Ahjin Studios' project greenlighting and budgeting decisions moving forward.

Conclusions and Recommendations

The data tells a decisive story: success in the film industry today isn't about outspending the majors, it's about **outmaneuvering** them. Ahjin Studios is not playing with billion-dollar IPs or global distribution muscle. But what it does have is agility, insight, and the ability to strike precisely where market openings align with audience appetite.

This analysis reveals key levers that Ahjin can pull to compete smartly and profitably.

1. Own Q4, Exploit Q1–Q2 White Space

Theatrical heavyweights concentrate around **November and December**, the golden window for prestige, family, and holiday fare. But **Q1 and Q2 remain underutilized**, offering space for breakout mid-tier hits.

What to do:

Anchor top-shelf releases in Q4 to ride seasonal audience momentum and awards buzz.

Slot mid-budget, concept-driven projects in Q1–Q2, where competition dips and discoverability spikes, especially for thrillers, dramas, or genre crossovers.

2. Runtime Isn't Cosmetic, It's Commercial

Films in the **150–180 minute** range, especially PG/PG-13, consistently earn more. These longer runtimes signal cinematic weight and draw audiences seeking immersive experiences.

What to do:

Embrace longer runtimes for stories with scale, emotional layering, or world-building.

Avoid sub-90-minute features unless stylistically intentional; they tend to underperform or be perceived as less valuable.

3. PG–PG13 is the Sweet Spot

Across the board, **PG and PG-13 ratings dominate box office returns**. They strike the balance between mass accessibility and creative freedom.

What to do:

Default development toward PG or PG-13 to maximize reach, especially internationally.

Reserve **R-rated material** for prestige plays or culturally urgent stories with clear ROI pathways.

4. Genre is Strategy, Not Just Aesthetic

Action and comedy still lead, but **thrillers, mysteries, and grounded dramas** quietly outperform when timed and budgeted well. These genres deliver big without requiring blockbuster spend.

What to do:

Focus on **genre hybrids**: e.g socially grounded thrillers, speculative dramas, sharp comedies with heart.

Defer high-VFX, blockbuster-style projects until the brand and budget can support them.

5. Budget Discipline Beats Budget Size

Big spenders flop just as often as they succeed. It's not about spending more, it's about spending **smart**.

What to do:

Create **tiered investment models** aligned with genre, scope, and expected return.

Bake data into the greenlight process, not after production, but at the script and planning stages.

Final Takeaways for Ahjin Studios

- **Prioritize Q4** for theatrical flagship releases; **strategically use Q1–Q2** for discovery plays.
- **Center storytelling around PG/PG-13 narratives** with emotional depth and thematic weight.
- **Use genre intentionally** not just creatively, but as part of a revenue model.
- **Match runtime and tone to audience expectations**, not just artistic impulses.
- And most importantly: **build smart, not cheap**.

With precision, timing, and a genre-savvy slate, Ahjin can build a commercially viable identity, one that's not reactive to the market but defines its own space within it.

